Robust efficiency of the Smears–Vohralík-type estimator for algebraically stabilised convection–diffusion–reaction

galerkit, gap 1 of litscan_2026-10.md. Estimator: `sv_estimator` (sv_convection_extension.md). 18 October 2026.

0. Summary and status

Results.

- **Theorem 1 (reliability).** The estimator is a guaranteed upper bound for E = |||e|||_⊕ + ‖S˜‖_*, for every σ ≥ 0 and every conforming u_h. The proof is complete and was given before; it is recalled here.
- **Theorem 2 (efficiency).** For each of the two functionals the estimator equilibrates, ℓ = Res_A and ℓ = −S˜:
  - the local contribution is bounded by the local dual norm of ℓ on a vertex neighbourhood, plus data oscillation;
  - the global estimator is bounded by ‖ℓ‖_* plus oscillation.
  - The constants depend only on d and the shape regularity. They are independent of ε, σ, b, c, h, the stabilisation tensor A, and u_h.
- **Corollary 3.** The estimator is two-sided and robust in E: E ≤ η_E ≤ C(1 + c_b)(E + osc).

What the proof relies on. It is complete given four cited results:

- the inverse inequalities and patch stability of [SV20, Lemmas 2.2 and 4.1];
- Verfürth's squeezed-bubble estimates [V05, Lemma 3.6];
- the stability of a patch-mean quasi-interpolant [V05, Lemma 3.3];
- an L^2-bounded commuting projection onto RT_0 with mixed boundary conditions [CW08; Licht 2019 — from memory, verify].

Only the last one is outside the papers already read. It is used in a single step (Lemma F), which can be bypassed when Γ_N = ∅ or σ = 0 (Remark 5.4).

Departures from the published arguments. The new ingredients are marked ◆ in the text:

1. **Weight-free construction (Lemma G).** The patch problems carry no SV weights; the weights enter only through the evaluation (Lemma A). The reaction-dominated case needs a new comparison argument: the best-approximation trick (Lemma G, case 1).
2. **Stabilisation as a diffusion tensor.** The stabilisation enters through D = εI + A. Compatibility is restored by an explicit defect shift, and its cost is bounded through a quasi-interpolant (Lemma H). This is what lets the bound cover inexact iterates.
3. **Augmented-norm constant (Remark 1.2).** The factor 3 in |||e|||_⊕ ≤ 3‖Res‖_* [EM16, Thm. 4.1] becomes 2 + max{1, c_b} for galerkit's norm. This was found while writing the proof, and the code is corrected (`kappa_aug`).

Not refereed. The result is checked numerically in §6.

1. Setting

Hypotheses

- **(H1)** Ω ⊂ $\mathbb{R}^d$ (d = 2, 3) is a polytope, and T_h is a conforming simplicial mesh with shape-regularity parameter ϑ.
- **(H2)** The coefficients satisfy:
  - ε > 0 is constant;
  - b ∈ W^{1,∞}(Ω)^d, c ∈ L^∞(Ω), f ∈ L^2(Ω), g_N ∈ L^2(Γ_N).
- **(H3)** There are σ ≥ 0 and c_b ≥ 0 with c − ½ div b ≥ σ and ‖c − ½ div b‖_∞ ≤ c_b σ. In particular, σ = 0 implies c = ½ div b.
- **(H4)** The Dirichlet part Γ_D has positive measure and contains the inflow boundary, so b·n ≥ 0 on Γ_N.
- **(H5)** u_D is the trace of a P1 function. Non-P1 data add the lifting terms of gap 4; their efficiency is not treated here.

- **(H6)** $u_h \in V_h = P1 \cap H^1(\Omega)$ with $u_h = u_D$ on Γ_D. The stabilisation is $S(u_h; u_h, v_h) = (A\nabla u_h, \nabla v_h)$ for all $v_h \in V_{h,0}$, with A cellwise constant, symmetric and positive semidefinite. This holds exactly for every algebraic stabilisation in galerkit (gap 2).
  - **Nothing is assumed about how u_h was obtained.** It may be any iterate.

Norms and functionals

Norms and weights.

- $|||v|||^2_\omega = \varepsilon\|\nabla v\|^2_\omega + \sigma\|v\|^2_\omega$.
- $\|\ell\|_{*,\omega} = \sup\{\ell(v) / |||v|||_\omega : v \in H^1_0(\omega) \setminus \{0\}\}$, and $\|\ell\|_* = \sup\{\ell(v) / |||v||| : v \in H^1_D(\Omega) \setminus \{0\}\}$.
- $\alpha_S = \min\{h_S \varepsilon^{-1/2}, \sigma^{-1/2}\}$ for an element or face S, with $\alpha_S = h_S \varepsilon^{-1/2}$ if $\sigma = 0$.
- $\theta_S = \varepsilon^{1/2} \sigma^{-1/2} / h_S$, so $\alpha_S = h_S \varepsilon^{-1/2} \min\{1, \theta_S\}$.

The residual functionals. For $v \in H^1_D$, with $e = u - u_h$:

- $\mathrm{Res}(v) = (f, v) + (g_N, v)_{\Gamma_N} - B(u_h, v) = B(e, v)$;
- $S^\sim(v) = (A\nabla u_h, \nabla v)$;
- $\mathrm{Res}_A = \mathrm{Res} - S^\sim$, so $\mathrm{Res}_A(v) = (r, v) - ((\varepsilon I + A)\nabla u_h, \nabla v) + (g_N, v)_{\Gamma_N}$, where $r = f - b\cdot\nabla u_h - c\, u_h$.

Both Res_A and $-S^\sim$ are of the generic form

$\ell(v) = (r_\ell, v) - (D\nabla u_h, \nabla v) + (g_\ell, v)_{\Gamma_N}$,

with $(r_\ell, D, g_\ell) = (r, \varepsilon I + A, g_N)$ for Res_A and $(0, A, 0)$ for $-S^\sim$.

Augmented norm. Write $B = B_S' + B_A'$ with

- $B_A'(w, v) = (b\cdot\nabla w + \tfrac{1}{2}(\mathrm{div}\, b)\, w, v)$;
- $B_S'(w, v) = \varepsilon(\nabla w, \nabla v) + ((c - \tfrac{1}{2}\,\mathrm{div}\, b)\, w, v)$.

Then $|||e|||_\oplus = |||e||| + \|B_A'(e, \cdot)\|_*$, and $E = |||e|||_\oplus + \|S^\sim\|_*$.

Remark 1.2 (◆ the factor in the augmented bound).

- By (H3), $|B_S'(w, v)| \le \varepsilon^{1/2}\|\nabla w\|\, \varepsilon^{1/2}\|\nabla v\| + c_b\, \sigma\|w\|\|v\| \le \max\{1, c_b\}\, |||w|||\, |||v|||$.
- Coercivity (H3)–(H4) gives $|||e||| \le \|\mathrm{Res}\|_*$, and $\|B_A'(e, \cdot)\|_* \le \|\mathrm{Res}\|_* + \max\{1, c_b\}|||e|||$.
- Hence $|||e|||_\oplus \le \kappa\|\mathrm{Res}\|_*$, with $\kappa = 2 + \max\{1, c_b\}$.
- EM16's $\kappa = 3$ is the case $c - \tfrac{1}{2}\,\mathrm{div}\, b \equiv \sigma$. All constant-coefficient examples in galerkit have $c_b = 1$. AJ27 Ex. 4.3 (c variable, $\sigma = \min c \approx 3\cdot10^{-21}$) does not, and its former augmented bound was not guaranteed.
- `abr17_estimator` and `sv_estimator` now use κ (`kappa_aug`). When σ does not control $c - \tfrac{1}{2}\,\mathrm{div}\, b$, they take the Friedrichs alternative $c_b = \|c - \tfrac{1}{2}\,\mathrm{div}\, b\|_\infty\, C_F^2/\varepsilon_{\min}$, but only if it lowers $\max\{1, c_b\}$. C_F is the Friedrichs constant of H^1_D, from `friedrichs_constant`:
  - **no Neumann boundary:** the H^1_0 constant of the bounding box, valid on any domain by zero extension;
  - **Neumann boundary on a box domain:** $C_F^{-2} = \Sigma_i \lambda_i$, with $\lambda_i = (\pi/L_i)^2$ if both faces normal to e_i are entirely Dirichlet, $(\pi/2L_i)^2$ if one is, and 0 otherwise. This follows by separation of variables, and counting a partly Neumann face as Neumann only lowers λ. For AJ27 Ex. 4.3 (Neumann at $x = 1$) it gives $C_F^2 = 1/(1.25\pi^2)$, and $\kappa \approx 8.1\cdot10^4$ is now guaranteed; the earlier $5\cdot10^4$ used the H^1_0 constant and was not;
  - **Neumann boundary on another domain:** no guaranteed constant is computed. For $\sigma > 0$ the σ-based c_b is kept. For $\sigma = 0$ the bounding-box constant is used and flagged as not guaranteed, also when it brings $\max\{1, c_b\}$ down to 1.

The same C_F enters η_{alg} and the Dirichlet-lifting terms.

- Conversely, $\|\mathrm{Res}\|_* \le \max\{1, c_b\}|||e||| + \|B_A'(e, \cdot)\|_* \le \max\{1, c_b\}|||e|||_\oplus$.

The estimator (as implemented)

For each vertex a, with patch ω_a and hat function ψ_a, define:

- V^a: RTN_1 on T_a, $H(\mathrm{div})$-conforming. Its normal component is:

  - 0 on ∂ω_a \ Γ_D (on all of ∂ω_a if a is interior);
  - −Π_F(ψ_a g_ℓ) on Neumann faces;
  - free on Dirichlet faces.
- g_a = Π₁(ψ_a r_ℓ − ∇ψ_a·D∇u_h).
- The defect δ_a = ℓ(ψ_a) if T_a has no free face ("compatible" patch), and δ_a = 0 otherwise. Set c_a = δ_a / |ω_a| and g˜_a = g_a − c_a.

Patch problems.

- σ > 0: σ^a = argmin over v ∈ V^a of J_a(v) = ε^{-1}‖ψ_a D∇u_h + v‖²_{ω_a} + σ^{-1}‖div v − g˜_a‖²_{ω_a}.
- σ = 0: the same flux term is minimised subject to div v = g˜_a. This is solvable, because the shift makes (g˜_a, 1)_{ω_a} consistent with the prescribed normal fluxes.

Notation. τ^a = ψ_a D∇u_h + σ^a and m^a = div σ^a − g˜_a, with m^a = 0 if σ = 0. Then σ_h = Σ_a σ^a and τ = D∇u_h + σ_h = Σ_a τ^a on each K.

Estimator.

- On each cell: η_{ℓ,K} = min{ε^{-1/2}‖τ‖_K, Σ_{F⊂∂K} c_F‖τ·n‖_F + m_K‖div τ‖_K} + σ^{-1/2}‖Σ_a m^a‖_K + m_K‖r_ℓ − Π₁r_ℓ‖_K, where
  - m_K = min{h_K/(π ε^{1/2}), σ^{-1/2}};
  - c_F² = (|F|/|K|) min{h_K²(1/π² + 2/(dπ))/ε, 1/σ + h_K/(d(σε)^{1/2})}.
- Total: η_ℓ = (Σ_K η²_{ℓ,K})^{1/2} + η_{ℓ,def}.
- The defect term is η_{ℓ,def} = min over s ∈ RT₀ with s·n = 0 on Γ_N of (ε^{-1}‖s‖² + σ^{-1}‖ψ_def − div s‖²)^{1/2}, with ψ_def = Σ_a c_a χ_{ω_a}. For σ = 0 the constraint div s = ψ_def is imposed instead.
- Then η_A = η_{Res_A}, η_S = min{gap 2/6 bound, η_{−S˜}}, and η_E = κ(η_A + η_S) + η_S, plus the boundary terms.

2. Results

Theorem 1 (reliability). Under (H1)–(H6):

- ‖ℓ‖_* ≤ η_ℓ + η_N(ℓ) for ℓ ∈ {Res_A, −S˜};
- hence |||e||| ≤ ‖Res‖_* ≤ η_A + η_S, |||e|||_⊕ ≤ κ(η_A + η_S), and E ≤ η_E.

Proof.

1. For any σ_h ∈ H(div) with σ_h·n = −Π_F g_ℓ on Γ_N: ℓ(v) = (r_ℓ − Π₁r_ℓ, v) + (Π₁r_ℓ − div σ_h, v) − (τ, ∇v) + (g_ℓ − Π_F g_ℓ, v)_{Γ_N}.
2. By construction Σ_a g˜_a = Π₁r_ℓ − ψ_def, so Π₁r_ℓ − div σ_h = ψ_def − Σ_a m^a.
3. Bound each term:
   - (τ, ∇v)_K: either Cauchy–Schwarz, or the identity (τ, ∇v)_K = (τ·n, v − v̄_K)_{∂K} − (div τ, v − v̄_K)_K together with the trace inequality ‖w‖²_F ≤ (|F|/|K|)(‖w‖² + (2/d) h_K ‖w‖‖∇w‖) and Poincaré;
   - (Σ_a m^a, v)_K ≤ σ^{-1/2}‖Σ_a m^a‖_K |||v|||_K;
   - the oscillation term, by Poincaré with min{h/(πε^{1/2}), σ^{-1/2}};
   - (ψ_def, v): write ψ_def = div s + z, then (ψ_def, v) = −(s, ∇v) + (z, v) ≤ (ε^{-1}‖s‖² + σ^{-1}‖z‖²)^{1/2} |||v|||.
4. Remark 1.2 gives the rest. ∎

Full details are in sv_convection_extension.md §4(i).

Theorem 2 (efficiency). Assume (H1)–(H6) and let ℓ ∈ {Res_A, −S˜}. Define:

- ω˜_K = the union of the patches ω_a with a ∈ V_K, and their neighbours;
- osc_K(ℓ) = α_K‖r_ℓ − Π₁r_ℓ‖_K + Σ_{F ⊂ ∂K ∩ Γ_N} ε^{-1/4}α_F^{1/2}‖g_ℓ − Π_F g_ℓ‖_F.

Then:

- **local:** η²_{ℓ,K} ≤ C Σ_{K' ⊂ ω˜_K} [‖ℓ‖²_{*,ω˜_{K'}} + osc_{K'}(ℓ)²] for every K;

- **global:** $\eta_\ell \le C(\|\ell\|_* + \mathrm{osc}(\ell))$.

The constant C depends only on d and ϑ. It is independent of ε, σ, b, c, h, A and u_h.

Corollary 3 (robust two-sided bound). Under (H1)–(H6):

- $\eta_A + \eta_S \le C(\|\mathrm{Res}_A\|_* + \|S^\sim\|_* + \mathrm{osc}) \le C(\max\{1, c_b\}|||e|||_\oplus + 2\|S^\sim\|_* + \mathrm{osc})$;
- hence $E \le \eta_E \le C'(1 + c_b)(E + \mathrm{osc})$.

Proof. Use $\|\mathrm{Res}_A\|_* \le \|\mathrm{Res}\|_* + \|S^\sim\|_*$, Remark 1.2, and the fact that η_S is at most the equilibrated bound $\eta_{-S^\sim}$. ∎

3. Tools

A1 (inverse inequalities) [SV20, Lemma 2.2]. For $\tau \in \mathrm{RTN}_1(K)$:

- $h_K^{1/2}\|\tau\cdot n\|_{\partial K} \le C_\partial \|\tau\|_K$;
- $h_K\|\mathrm{div}\,\tau\|_K \le C_{\mathrm{div}}\|\tau\|_K$.

Both constants are explicit in d and ϑ.

A2 (patch stability) [SV20, Lemma 4.1]. Let $g \in P_1(T_a)$ and let τ be a broken RTN_1 field on T_a, with the compatibility $(g, 1)_{\omega_a} = \langle \text{prescribed normal flux} \rangle$ on compatible patches. Then

$$\min\{\|\tau + v\|_{\omega_a} : v \in V^a,\ \mathrm{div}\, v = g\} \le C \sup\{(g, w)_{\omega_a} - (\tau, \nabla w)_{\omega_a} : w \in H^1_*(\omega_a),\ \|\nabla w\| = 1\}.$$

Here H^1_* consists of the functions with mean zero on ω_a (compatible patches), or vanishing on $\partial\omega_a \cap \Gamma_D$. With Neumann faces, the version used is the standard affine shift of the prescribed normal flux [EV15]; not re-proved.

A3 (bubbles) [V05, Lemma 3.6]. With b_K the element bubble and b_F^θ the face bubble squeezed by $\theta_F = \min\{\varepsilon^{1/2}\alpha_F/h_F, 1\}$:

- $(w, b_K w)_K \ge c\|w\|^2_K$ and $|||b_K w|||_K \le C\,\alpha_K^{-1}\|w\|_K$, for $w \in P_1(K)$;
- $(\mu, b_F^\theta \mu)_F \ge c\|\mu\|^2_F$, $|||b_F^\theta \mu|||_{\omega_F} \le C\,\varepsilon^{1/4}\alpha_F^{-1/2}\|\mu\|_F$, and $\|b_F^\theta \mu\|_{\omega_F} \le C\,\varepsilon^{1/4}\alpha_F^{1/2}\|\mu\|_F$, for $\mu \in P_1(F)$.

A4 (quasi-interpolant) [variant of V05, Lemma 3.3]. Let $I_c v = \Sigma_{a \text{ compatible}} \bar v_{\omega_a} \psi_a$, where $\bar v_\omega$ is the mean over ω. Then $|||I_c v||| \le C_I\,|||v|||$ for $v \in H^1_D$.

- **The σ-part** follows from $|\bar v_a| \le |\omega_a|^{-1/2}\|v\|_{\omega_a}$.
- **The ε-part.** On K, $\nabla I_c v = \Sigma_{a\in V_K} (\bar v_a - \bar v_K)\nabla\psi_a - \Sigma_{b \text{ free}} \bar v_b \nabla\psi_b$.
  - The first sum is bounded by Poincaré on the vertex patches of K.
  - The second sum runs over the excluded vertices b. Each such patch has a face on Γ_D, where $v = 0$. Friedrichs gives $|\bar v_b| \le C\,h_b^{1-d/2}\|\nabla v\|_{\omega_b}$.
- **Comparison with [V05].** Verfürth's set excludes the vertices on Γ_D. Here, in addition, vertices whose patch has an opposite face on Γ_D are excluded; the same argument applies to them.

A5 (commuting projection) [Christiansen–Winther 2008; Licht 2019, mixed boundary conditions — from memory, verify]. There is $\Pi_h : H(\mathrm{div}) \cap \{s\cdot n = 0 \text{ on } \Gamma_N\} \to \mathrm{RT}_0$ with the same boundary condition, such that:

- $\|\Pi_h s\| \le C_\Pi \|s\|$;
- $\mathrm{div}\,\Pi_h s = \Pi_0\,\mathrm{div}\, s$.

A6 (finite overlap). If $\omega_1, \dots, \omega_k \subset \Omega$ cover each point at most N times, then $\Sigma_j \|\ell\|^2_{*,\omega_j} \le N\,\|\ell\|^2_*$.

- Proof: pick $v_j \in H^1_0(\omega_j)$ with $|||v_j|||_{\omega_j} = 1$ and $\ell(v_j) = \|\ell\|_{*,\omega_j}$ (up to a factor $1 - \delta$), and set $t_j = \ell(v_j)$ and $v = \Sigma_j t_j v_j$.
- Then $\ell(v) = \Sigma_j t_j^2$, while $|||v|||^2 \le N\,\Sigma_j t_j^2\,|||v_j|||^2 = N\,\Sigma_j t_j^2$.
- So $\|\ell\|_* \ge (\Sigma_j t_j^2)^{1/2} / \sqrt{N}$. ∎

4. Lemmas

Lemma A (◆ the evaluated flux term carries the SV weight). For $\tau \in \mathrm{RTN}_1(K)$ and $\omega_K = \min\{1, \theta_K^{1/2}\}$:

min{ε^{-1/2}‖τ‖_K, Σ_F c_F‖τ·n‖_F + m_K‖div τ‖_K} ≤ C_A ε^{-1/2} ω_K ‖τ‖_K.

Proof.

1. If θ_K ≥ 1, the first entry of the min gives the bound with C_A = 1.
2. If θ_K < 1, then m_K = σ^{-1/2} = (h_K/ε^{1/2}) θ_K.
3. Also 1/σ + h_K/(d(σε)^{1/2}) = (h_K²/ε)(θ_K² + θ_K/d) ≤ 2(h_K²/ε)θ_K, and |F|/|K| ≤ C_ϑ h_K^{-1}. So c_F ≤ C h_K^{1/2} ε^{-1/2} θ_K^{1/2}.
4. With A1: Σ_F c_F‖τ·n‖_F + m_K‖div τ‖_K ≤ C ε^{-1/2}(θ_K^{1/2} C_∂ + θ_K C_div)‖τ‖_K ≤ C_A ε^{-1/2} θ_K^{1/2}‖τ‖_K. ∎

Lemma B (conforming reconstruction). For each vertex a there is v° ∈ V^a with

‖ψ_a Dᵛu_h + v°‖_{ω_a} ≤ C_B h_a^{1/2} J_a,

where J_a² = Σ_{F ∈ F_a} ‖j_F‖²_F, the sum runs over the faces F containing a, and

- j_F = [[Dᵛu_h·n]] on interior faces;
- j_F = Dᵛu_h·n − Π_F g_ℓ on Neumann faces.

Proof.

1. Define v° by its RTN₁ degrees of freedom:
   - interior moments: those of −ψ_a Dᵛu_h on each K;
   - normal moments on interior faces of T_a: the average of the two traces of −ψ_a Dᵛu_h·n;
   - normal moments on ∂ω_a: 0 (there ψ_a = 0), or the prescribed −Π_F(ψ_a g_ℓ) on Neumann faces, or the one-sided trace on Dirichlet faces.
2. Then w = ψ_a Dᵛu_h + v° is a broken RTN₁ field.
   - Its interior degrees of freedom vanish.
   - Its face moments are ±½ψ_a j_F on interior faces and ψ_a j_F on Neumann faces, and 0 otherwise. These are exact, because ψ_a Dᵛu_h·n ∈ P₁(F).
3. Scaling to the reference element gives ‖w‖_K ≤ C h_K^{1/2} Σ_{F⊂∂K} ‖ψ_a j_F‖_F. Use ‖ψ_a‖_∞ = 1. ∎

Lemma G (◆ patch lemma, weight-free construction). Let RES_a² = Σ_{K∈T_a} α_K²‖r_ℓ‖²_K + Σ_{F∈F_a} ε^{-1/2} α_F ‖j_F‖²_F, and let θ_a = ε^{1/2}σ^{-1/2}/h_a. Then

ε^{-1}ω_a²‖τ^a‖²_{ω_a} + σ^{-1}‖m^a‖²_{ω_a} ≤ C(RES_a² + [θ_a < 1] σ^{-1} c_a² |ω_a|).

For σ = 0 the second term on the left and the bracket are absent.

Proof, case 2 (θ_a ≥ 1, or σ = 0: diffusion regime). Here ω_a = 1.

1. By A2 there is v* ∈ V^a with div v* = g̃_a. The shift makes the data compatible, and the constant c_a drops out against mean-zero test functions.
2. The flux satisfies ‖ψ_a Dᵛu_h + v*‖ ≤ C sup over w ∈ H¹_* with ‖∇w‖ = 1 of [(Π₁(ψ_a r_ℓ), w) + Σ_F (ψ_a j_F, w)_F].
3. This follows from elementwise integration by parts, using div(ψ_a Dᵛu_h)|_K = ∇ψ_a·Dᵛu_h (Dᵛu_h is constant on K).
4. Poincaré on ω_a (Friedrichs on Dirichlet patches) and the trace inequality give ‖ψ_a Dᵛu_h + v*‖ ≤ C(h_a‖r_ℓ‖_{ω_a} + h_a^{1/2}J_a).
5. Hence J_a(σ^a) ≤ J_a(v*) = ε^{-1}‖ψ_a Dᵛu_h + v*‖² ≤ C(ε^{-1}h_a²‖r_ℓ‖² + ε^{-1}h_a J_a²).
6. Since θ ≥ 1, α_K ≃ h_a ε^{-1/2} and α_F ≃ h_a ε^{-1/2}. So the right-hand side is ≤ C RES_a², and the left-hand side of the lemma is ≤ J_a(σ^a).
7. For σ = 0 the same competitor is admissible in the constrained problem. ∎

Proof, case 1 (θ_a < 1: reaction regime; ◆ the best-approximation trick).

1. **Why the SV competitor does not suffice.** Its flux part costs ε^{-1}h_aJ_a², which is θ_a^{-1} times larger than the efficient size ε^{-1/2}σ^{-1/2}J_a². SV20 compensate with the weight w_a² ≃ θ_a in the construction. Here the weight is only in the evaluation.
2. **The comparison flux.** Let v_b = argmin over v ∈ V^a of ‖ψ_a Dᵛu_h + v‖, with no divergence condition, and let δ = ‖ψ_a Dᵛu_h + v_b‖ (this δ is an approximation error, not the defect δ_a). Since σ^a

$\in V^a$, $\|\tau^a\| \geq \delta$. Optimality $J_a(\sigma^a) \leq J_a(v_b)$ gives $\varepsilon^{-1}\|\tau^a\|^2 + \sigma^{-1}\|m^a\|^2 \leq \varepsilon^{-1}\delta^2 + \sigma^{-1}\|m_b\|^2$, with $m_b = \operatorname{div} v_b - g\tilde{}_a$.

3. **Equilibration term:** $\sigma^{-1}\|m^a\|^2 \leq \sigma^{-1}\|m_b\|^2$. This uses $\delta \leq \|\tau^a\|$.
4. **Bound on m_b.**
   - Write $m_b = \operatorname{div}_h(\psi_a D\nabla u_h + v_b) - \Pi_1(\psi_a r_\ell) + c_a$.
   - A1 gives $\|\operatorname{div}_h(\ldots)\| \leq C h_a^{-1}\delta$, and Lemma B gives $\delta \leq \|\psi_a D\nabla u_h + v^\circ\| \leq C h_a^{1/2} J_a$.
   - So $\sigma^{-1}\|m_b\|^2 \leq C(\sigma^{-1}h_a^{-1}J_a^2 + \sigma^{-1}\|r_\ell\|^2_{\omega_a} + \sigma^{-1}c_a^2|\omega_a|)$.
   - Since $\theta_a < 1$, $\alpha_F \simeq \sigma^{-1/2}$ and $\alpha_K \simeq \sigma^{-1/2}$. So $\sigma^{-1}h_a^{-1}J_a^2 = \theta_a \varepsilon^{-1/2}\sigma^{-1/2}J_a^2 \leq C\,\varepsilon^{-1/2}\alpha_F J_a^2$, and $\sigma^{-1}\|r_\ell\|^2 \leq C\,\Sigma\,\alpha_K^2\|r_\ell\|^2$.
5. **Flux term.**
   - From step 2, $\|\tau^a\|^2 \leq \delta^2 + (\varepsilon/\sigma)\|m_b\|^2$.
   - Multiply by $\varepsilon^{-1}\omega_a^2 = \varepsilon^{-1}\theta_a$: $\varepsilon^{-1}\theta_a\|\tau^a\|^2 \leq \theta_a\,\varepsilon^{-1}\delta^2 + \theta_a\,\sigma^{-1}\|m_b\|^2$.
   - The first term is $\leq C\,\theta_a\,\varepsilon^{-1}h_a J_a^2 = C\,\varepsilon^{-1/2}\sigma^{-1/2}J_a^2$. The second is θ_a times the bound of step 4.
6. Both terms are therefore $\leq C(\mathrm{RES}_a^2 + \sigma^{-1}c_a^2|\omega_a|)$. ∎

Lemma H (defect terms).

(i) **Local defect term.** For $\theta_a < 1$: $\sigma^{-1}c_a^2|\omega_a| \leq C\|\ell\|^2_{*,\omega_a}$.

- Proof: $c_a|\omega_a| = \ell(\psi_a) \leq \|\ell\|_{*,\omega_a}|||\psi_a|||_{\omega_a}$, and $|||\psi_a|||^2_{\omega_a} \leq C(\varepsilon h_a^{-2} + \sigma)|\omega_a| = C\sigma(\theta_a^2 + 1)|\omega_a| \leq 2C\sigma|\omega_a|$. ∎

(ii) **Global defect term** (◆). The functional $\ell_{\mathrm{def}}(v) = (\psi_{\mathrm{def}}, v)$ satisfies $\|\ell_{\mathrm{def}}\|_* \leq C_I\|\ell\|_*$. If A5 holds, then also $\eta_{\ell,\mathrm{def}} \leq C_\Pi C_I \|\ell\|_*$.

- Proof.
  1. $\ell_{\mathrm{def}}(v) = \Sigma_{a \text{ compatible}} c_a(\chi_{\omega_a}, v) = \Sigma_a \ell(\psi_a)\,\bar{v}_{\omega_a} = \ell(I_c v)$. A4 gives the first claim.
  2. For the second, let (s, z) attain the continuous dual norm: $\psi_{\mathrm{def}} = \operatorname{div} s + z$, with $\|\ell_{\mathrm{def}}\|^2_* = \varepsilon^{-1}\|s\|^2 + \sigma^{-1}\|z\|^2$. This is the duality for the symmetric operator $-\varepsilon\Delta + \sigma$, and $z = 0$ if $\sigma = 0$.
  3. Then $s_h = \Pi_h s$ is admissible for the discrete minimum, because $\operatorname{div} s_h = \Pi_0 \operatorname{div} s = \psi_{\mathrm{def}} - \Pi_0 z$ ($\psi_{\mathrm{def}} \in P_0$).
  4. Hence $\eta^2_{\ell,\mathrm{def}} \leq \varepsilon^{-1}C_\Pi^2\|s\|^2 + \sigma^{-1}\|z\|^2 \leq C_\Pi^2\|\ell_{\mathrm{def}}\|^2_*$. ∎

(iii) **For ℓ = Res_A,** $\ell(\psi_a)$ is minus the algebraic residual at a. So both defect terms vanish at the discrete solution, and at an iterate they are bounded by $\|\mathrm{Res_A}\|_*$. This makes Theorem 2 hold for inexact iterates.

Lemma D (residual estimator ≤ local dual norm) [V05 proof of Thm. 4.1, with A3]. For every K and every face F:

- $\alpha_K\|\Pi_1 r_\ell\|_K \leq C(\|\ell\|_{*,K} + \alpha_K\|r_\ell - \Pi_1 r_\ell\|_K)$;
- $\varepsilon^{-1/4}\alpha_F^{1/2}\|j_F\|_F \leq C(\|\ell\|_{*,\omega_F} + \alpha_F\|r_\ell\|_{\omega_F} + \text{Neumann data oscillation})$.

Proof.

1. **Element term.** $\ell(b_K w) = (r_\ell, b_K w)_K$ for $w \in P_1(K)$. The diffusion term vanishes, because $b_K = 0$ on ∂K and $\operatorname{div}(D\nabla u_h) = 0$ in K. Take $w = \Pi_1 r_\ell$ and apply A3.
2. **Face term.** $\ell(b_F^\theta \mu) = (r_\ell, b_F^\theta \mu)_{\omega_F} + (j_F, b_F^\theta \mu)_F$. Take $\mu = j_F$, apply A3, and use the element bound.
3. Only the form of ℓ is used; the bilinear form B and u_h's equation are not. ∎

5. Proof of Theorem 2

1. **Lemma A on each cell.** On K, $\tau = \Sigma_{a\in V_K} \tau^a$. With $\omega_K \simeq \omega_a$ for $a \in V_K$ (shape regularity): $\eta_{\ell,K} \leq C\,\Sigma_{a\in V_K}[\varepsilon^{-1/2}\omega_a\|\tau^a\|_K + \sigma^{-1/2}\|m^a\|_K] + m_K\|r_\ell - \Pi_1 r_\ell\|_K$. The last term uses $m_K \leq \alpha_K$.
2. **Lemma G for each a,** then Lemma H(i) for the local defect: $\eta^2_{\ell,K} \leq C\,\Sigma_{a\in V_K}(\mathrm{RES}_a^2 + \|\ell\|^2_{*,\omega_a}) + \mathrm{osc}_K^2$.

3. **Lemma D,** with $\|r_\ell\| \le \|\Pi_1 r_\ell\| + \|r_\ell - \Pi_1 r_\ell\|$: $\mathrm{RES}_a^2 \le C \sum_{K' \subset \omega_a, F \in \mathcal{F}_a} (\|\ell\|^2_{\omega_F} + \|\ell\|^2_{K'} + \mathrm{osc}^2_{K'})$. This is the local bound.
4. **Summing over K.** The sets K, ω_F and ω_a overlap at most $N(d, \vartheta)$ times. A6 gives $\sum_K \eta^2_{\ell,K} \le C(\|\ell\|^2_* + \mathrm{osc}^2)$. Adding Lemma H(ii) for $\eta_{\ell,\mathrm{def}}$ gives the global bound. ∎

Remark 5.1 (where robustness comes from). Every constant above depends only on d and ϑ. The parameters ε, σ, b and c enter only through:

- the weights α and θ, which cancel exactly in Lemmas A, G and H(i);
- the functional ℓ itself (through D, r_ℓ and g_ℓ), which is never bounded by the coefficients.

The stabilisation tensor A appears only inside ℓ (as D). It therefore does not affect the constants, whatever the limiter.

Remark 5.2 (local versus global). Theorem 2 is local in the dual norms of ℓ. The conversion to the error is local only for $|||e|||_{\tilde{\omega}_K}$. The convective part $\|B_{A'}(e, \cdot)\|_{*,\tilde{\omega}_K}$ is a localised dual norm, as in [V05]. Global efficiency in E is Corollary 3.

Remark 5.3 (the direct version). The `direct` variant ($D = \varepsilon I$, $\sigma > 0$) is not compatible in general: $\ell(\psi_a) = \mathrm{Res}(\psi_a) = S(u_h; u_h, \psi_a)$ at the discrete solution. Its efficiency would carry the consistency term $\sum_a \sigma^{-1} S(u_h; u_h, \psi_a)^2 / |\omega_a|$. Corollary 3 does not need it, because galerkit uses $\min\{\eta_{\mathrm{dir}}, \eta_A + \eta_S\} \le \eta_A + \eta_S$.

Remark 5.4 (where A5 is needed). A5 is used only in Lemma H(ii), the quasi-optimality of the global RT_0 defect bound. That covers both $\sigma > 0$ and $\sigma = 0$, where $z = 0$. The discrete inf-sup condition alone gives $\|s_h\|_{H(\mathrm{div})} \lesssim \|\psi_{\mathrm{def}}\|$, which is not $\|s_h\| \lesssim \|\psi_{\mathrm{def}}\|_{H^{-1}}$.

- **$\Gamma_N = \emptyset$:** the space has no boundary condition, and the smoothed projection of [CW08] suffices.
- **Mixed boundary conditions:** these need the version of Licht (2019), as far as I recall; unverified.
- **What it affects:** only the defect terms, which vanish for $\ell = \mathrm{Res}_A$ at the discrete solution. Without A5, Theorem 2 still holds for $\ell = \mathrm{Res}_A$ at the discrete solution. For $\ell = -\tilde{S}$, the defect term then has only the upper bound $\|\ell_{\mathrm{def}}\|_* \le C_I \|\tilde{S}\|_*$ for the true dual norm, not for its RT_0 computation.

6. Numerical verification (smoke runs)

Lemmas A and G with Prop. 2 ($\eta \lesssim$ local residual estimator). For the outflow-layers problem, both functionals, BJK and SMUAS, on uniform meshes:

	quantity	max over $\sigma \in \{0, 10^{-2}, 1\}$, $\varepsilon = 10^{-2} \ldots 10^{-8}$, $n \in \{8, 32\}$
$\ell = \mathrm{Res}_A$	$\max_K \eta^2_{\ell,K} / \sum_{K' \sim K} (\mathrm{RES}^2_{K'} + \mathrm{osc}^2_{K'})$	0.63
$\ell = \mathrm{Res}_A$	$\sum_K \eta^2_{\ell,K} / \sum_K (\mathrm{RES}^2_K + \mathrm{osc}^2_K)$	2.39
$\ell = -\tilde{S}$	$\max_K$	0.56
$\ell = -\tilde{S}$	global	2.36

- **No growth in ε.** The ratios rise from about 0.03 (diffusion regime) to about 2.4 as the reaction regime sets in, and then stay bounded.
- **Lemma D for $\ell = -\tilde{S}$.** Here a reference is available: the P2 Riesz representative, a lower bound of $\|\tilde{S}\|_*$. With it, $\mathrm{RES}/\|\tilde{S}\|_* = 2.4$–$2.5$ for $\sigma = 0$ and $\varepsilon = 10^{-2} \ldots 10^{-8}$.
  - For $\sigma = 1$ the ratio grows (3.9 at $\varepsilon = 10^{-6}$, 12 at 10^{-8}). This is the known reference artifact: $\|\tilde{S}\|_*$ concentrates in layers of width $\sqrt{(\varepsilon/\sigma)}$ that the same-mesh P2 space cannot resolve, as the nested-mesh check of gap 6 showed.
- **Corollary 3.** The effectivities of `sv` in E are 2.9–3.8 for $\sigma = 0$ and 3.2–10 for $\sigma = 1$, against same-mesh references (§9 of the summary). The upper end for $\sigma = 1$ is the same artifact.

Scripts. `scratchpad/g9.py` (ratios) and `tests/test_cdr_sv_estimator.py` (bounds).

7. What is not covered

- **Variable ε.** Reliability and local efficiency are now in §11. Robustness in coefficient jumps is shown only numerically, and only for quasi-monotone ε.
- **Non-P1 Dirichlet data.** The efficiency of the gap 4 lifting terms is not treated.

- **Neumann faces in A2.** The affine-shift version of SV20 Lemma 4.1 is cited, not re-proved.
- **A5 with mixed boundary conditions for σ > 0.** The citation is unverified.
- **The dependence on c_b in Corollary 3.** It is unavoidable with this norm (Remark 1.2). For c_b ≫ 1 one should take the energy norm with the full reaction c − ½ div b, which galerkit's σ only bounds from below.
- **The direct variant (Remark 5.3)** and the gap 6 hybrid have no efficiency claim.

8. References

- **[SV20]** I. Smears, M. Vohralík, ESAIM: M2AN 54 (2020) 1951–1973. Lemmas 2.2, 4.1, 4.2; Theorem 4.4. [read]
- **[V05]** R. Verfürth, SIAM J. Numer. Anal. 43 (2005) 1766–1782. Lemmas 3.1, 3.3, 3.6; Theorem 4.1. [read]
- **[EM16]** M. Eigel, C. Merdon, J. Sci. Comput. 67 (2016). Theorem 4.1. [read]
- **[EV15]** A. Ern, M. Vohralík, SIAM J. Numer. Anal. 53 (2015) 1058–1081: polynomial-degree-robust patch stability. [from memory — verify]
- **[CW08]** S. H. Christiansen, R. Winther, Math. Comp. 77 (2008) 813–829: smoothed projections. [from memory — verify]
- **[Licht 2019]** M. W. Licht, Smoothed projections and mixed boundary conditions, Math. Comp. 88 (2019) 607–635. [from memory — verify]
- **[AV11]** M. Ainsworth, T. Vejchodský, Numer. Math. 119 (2011): reaction–diffusion with variable coefficients. [from memory — verify]

9. Addendum: local efficiency in a localisable error measure (gap 4 of the scan)

The obstacle and the answer

The obstacle. Theorem 2 is local in the dual norms of the residual functionals. Converting them to the error brings in $\|B_A'(e, \cdot)\|_*$, and this global dual norm is not a sum of local contributions.

The answer. Everything in E localises except one term. That term depends only on u_h, and the estimator already computes it. The tool is the localisation of dual norms of functionals that vanish on a partition of unity [BMV].

Definitions

Local dual norm. For each vertex a, $H^1_*(\omega_a) = \{v \in H^1(\omega_a) : v = 0 \text{ on } \partial\omega_a \setminus \Gamma_N\}$, and $\|\ell\|_{,a} = \sup\{\ell(v)/|||v|||_{\omega_a} : v \in H^1_*(\omega_a)\}$.

Localised error measure.

$E_{loc}^2 = |||e|||^2 + \Sigma_a (\|B_A'(e, \cdot)\|^2_{,a} + \|S^\sim\|^2_{,a})$.

The local form is $E_{loc}(\omega)^2$, with the sums restricted to the patches in ω. E_{loc} is a sum of local contributions by construction. Its first two parts form the localised augmented norm $|||e|||^2_{\oplus,loc} = |||e|||^2 + \Sigma_a \|B_A'(e, \cdot)\|^2_{*,a}$.

Nodal moment functionals. For $\ell \in \{Res_A, S^\sim\}$, $\ell_{def}(v) = \Sigma_{a \text{ compatible}} \ell(\psi_a)\, \bar{v}_{\omega_a}$.

- For $S^\sim$, $\ell(\psi_a) = S(u_h; u_h, \psi_a)$: the nodal moments of the stabilisation.
- For Res_A, $\ell(\psi_a)$ is minus the algebraic residual, which is zero at the discrete solution.
- Their dual norms are bounded above by the computed defect terms $\eta_{def,S}$ and $\eta_{def,A}$ (Theorem 1, step 3). With A5 they are also bounded below by them (Lemma H(ii)).

Results

Lemma L (robust localisation). For $\ell \in \{Res_A, S^\sim\}$,

$\|\ell\|_* \le C_L (\Sigma_a \|\ell\|^2_{,a})^{1/2} + \|\ell_{def}\|_$, and $\Sigma_a \|\ell\|^2_{,a} \le (d+1)\|\ell\|^2_$.

C_L depends only on d and ϑ. It does not depend on ε or σ.

Proof.

1. Let v ∈ H¹_D. Set v̄_a = mean over ω_a of v for compatible a, and v̄_a = 0 for patches with a free (Dirichlet) face. Since Σ_a ψ_a = 1, $\ell(v) = \Sigma_a \ell(\psi_a(v - \bar{v}_a)) + \Sigma_{a \text{ compatible}} \bar{v}_a \ell(\psi_a) = \Sigma_a \ell(\psi_a(v - \bar{v}_a)) + \ell_{\text{def}}(v)$, and ψ_a(v − v̄_a) ∈ H¹_*(ω_a).
2. **The robust bound $|||\psi_a(v - \bar{v}_a)|||_{\omega_a} \le C|||v|||_{\omega_a}$.**
   - Diffusion part: $\varepsilon\|\nabla(\psi_a w)\|^2 \le 2\varepsilon(\|\nabla w\|^2 + \|\nabla\psi_a\|^2_\infty\|w\|^2)$, with w = v − v̄_a. Poincaré (compatible patches) or Friedrichs (patches with a Dirichlet face) gives ‖w‖ ≤ C h_a‖∇v‖_{ω_a}, and ‖∇ψ_a‖_∞ ≤ C h_a^{-1}.
   - Reaction part: $\sigma\|\psi_a w\|^2 \le 2\sigma(\|v\|^2 + |\omega_a|\,\bar{v}_a^2) \le 4\sigma\|v\|^2_{\omega_a}$.
3. Hence $|\Sigma_a \ell(\psi_a(v - \bar{v}_a))| \le C(\Sigma_a \|\ell\|^2_{,a})^{1/2}(\Sigma_a |||v|||^2_{\omega_a})^{1/2} \le C\sqrt{(d+1)}(\Sigma_a \|\ell\|^2_{,a})^{1/2}|||v|||$.
4. The second inequality is A6 with N = d+1, the maximal number of vertex patches covering a point. ∎

Theorem 4 (localisable two-sided bounds). Under (H1)–(H6):

1. **Upper bound, guaranteed:** E_loc ≤ √(d+1) E ≤ √(d+1) η_E (`eta_Eloc`).
2. **Local efficiency, robust:** $\eta_{A,K} + \eta^{\text{patch}}_{S,K} \le C(1 + c_b)(E_{\text{loc}}(\tilde{\omega}_K) + \text{osc}(\tilde{\omega}_K))$. Here $\eta^{\text{patch}}_{S,K}$ is the cellwise part, without the global defect term.
3. **Equivalence modulo computable terms:** $E \le C(1 + c_b)(E_{\text{loc}} + \eta_{\text{def},S} + \eta_{\text{def},A})$. At the discrete solution, $\eta_{\text{def},A} = 0$.

Together: $E_{\text{loc}} \lesssim E \lesssim E_{\text{loc}} + \eta_{\text{def},S}$. In the global error measure, the only non-localisable part is the dual norm of the stabilisation's nodal moments. It is a function of u_h alone, and the estimator computes it.

Proof.

1. **Part 1** is Lemma L (second inequality) applied to $\ell = B_A'(e, \cdot)$ and $\ell = S^\sim$, plus Theorem 1. The functional B_A'(e, ·) has no partition-of-unity property, so only this direction holds for it.
2. **Part 2.** The local bounds of Theorem 2 use $\|\ell\|_{,K}$ and $\|\ell\|_{,\omega_F}$, both of which are ≤ $\|\ell\|_{*,a}$ for a ∈ K or a ∈ F. Then:
   - for ℓ = Res_A: $\|\text{Res_A}\|_{,a} \le \|\text{Res}\|_{,a} + \|S^\sim\|_{*,a}$;
   - $\|\text{Res}\|_{,a} \le \max\{1, c_b\}|||e|||_{\omega_a} + \|B_A'(e, \cdot)\|_{,a}$, because B = B_S' + B_A' and $|B_S'(e, v)| \le \max\{1, c_b\}|||e|||_{\omega_a}|||v|||_{\omega_a}$ for v ∈ H¹_*(ω_a);
   - for $\ell = S^\sim$, the local defect term of Lemma H(i) is ≤ $C\|S^\sim\|_{*,a}$.
3. **Part 3.**
   - First, $E \le \kappa\|\text{Res}\|_* + \|S^\sim\|_* \le \kappa\|\text{Res_A}\|_* + (\kappa + 1)\|S^\sim\|_*$.
   - Apply the first inequality of Lemma L to Res_A and to $S^\sim$, then bound $\|\text{Res_A}\|_{*,a}$ as in part 2.
   - Finally use $\|\ell_{\text{def}}\|_* \le \eta_{\text{def}}$. This direction needs no A5. ∎

Remarks

Remark 9.1 (is E_loc weaker than E?). Yes, by part 1.

- **Example where it is much weaker.** Take B_A'(e, ·) smooth and of global extent, with σ = 0. Then $\|B_A'(e, \cdot)\|_* \sim \varepsilon^{-1/2}\|b\cdot\nabla e\|_{H^{-1}}$, while $\Sigma_a \|\cdot\|^2_{*,a} \sim \varepsilon^{-1}h^2\|b\cdot\nabla e\|^2$.
- **Why E is still controlled.** Part 3 says this can only happen when $\|\text{Res_A}\|_*$ and $\|S^\sim\|_*$ are themselves globally large. For Res_A that is excluded by Lemma L. So it can only come from the stabilisation's nodal moments, i.e. $\eta_{\text{def},S}$, which is computed.
- **In the smoke runs,** $\eta_{\text{def},S}/E_{\text{loc}}$ = 0.08–0.54 and E_{loc}/E = 0.48–1.10.

Remark 9.2 (literature).

- The localisation argument is that of [BMV] ($W^{-1,q}$, partition of unity), in a robust weighted form.
- Verfürth [V05] and Cai & Cai prove efficiency only globally in the augmented norm. Sangalli (Math. Comp. 2008) uses a different, almost-robust norm (1D).
- I found no result that localises the augmented norm for stabilised schemes, or that isolates the stabilisation's nodal moments as the only global term. This is based on searches of abstracts only.

Numerics (smoke runs; same-mesh patch P2 references; `localized_error`)

	$\sigma = 0$, $\varepsilon = 10^{-2} \dots 10^{-8}$	$\sigma = 1$, $\varepsilon = 10^{-2} \dots 10^{-8}$
E_{loc} / E	0.48–1.05	0.52–1.10
η_E / E_{loc}	3.6–5.9	3.1–11
$\max_K (\eta^2_{A,K} + \eta^2_{S,K}) / E_{loc}(\tilde{\omega}_K)^2$	0.09–0.64	0.19–49

- **$\sigma = 0$:** local efficiency holds, with ratios ≤ 0.64, flat in ε.
- **$\sigma = 1$, small ε:** the large local ratios are the unresolved reference. Recomputing the patch dual norms on k-times refined patches (outflow layers, $\varepsilon = 10^{-6}$, $n = 8$) lowers the ratio from 4.81 ($k = 1$) to 3.00, 1.85 and 1.14 ($k = 2, 4, 8$). For $\varepsilon = 10^{-8}$ it falls from 49 to 9.4 at $k = 8$, still decreasing; layers of width 10^{-4} are not yet resolved there. For $\sigma = 0$ the refined reference converges, at 0.09.
- **Tests:** `tests/test_cdr_sv_estimator.py::test_localized_measure`.

Reference added.

- [BMV] J. Blechta, J. Málek, M. Vohralík, Localization of the W^{-1,q} norm for local a posteriori efficiency, HAL hal-01332481; published in IMA J. Numer. Anal. 40 (2020) [from memory — verify].

10. Addendum: adaptive stopping of the nonlinear iteration (item 6 of the scan)

Setting. u^i is any iterate of any solver for the algebraically stabilised system: fixed point, Newton, or a truncated linear solve. Write E(u^i) for the measure E evaluated at u^i, with $S^{\sim}$ the stabilisation functional of the iterate.

The split. At an iterate, $\mathrm{Res}_A(\psi_a) = -r_a$, where r is the algebraic residual vector. The A-version moves r into the defect shift, so the A-version bound splits as $\kappa\eta_A + \eta_S + bd = \eta_{disc} + \eta_{alg}$:

- $\eta_{alg} := \kappa\, \eta_{def,A}(u^i)$, the RT_0 bound of the spread defect $\psi_{def} = \Sigma_a \delta_a \chi_{\omega_a}/|\omega_a|$, computed by `sv_algebraic` from r alone;
- $\eta_{disc} := \kappa(\eta_A - \eta_{def,A}) + \eta_S + bd$.

Proposition 5.

1. **Reliability at every iterate:** $E(u^i) \leq \eta_E(u^i) \leq \eta_{disc} + \eta_{alg}$. No assumption is made on the solver.
2. **At the stopped iterate:** if the iteration stops at $\eta_{alg} \leq \gamma\, \eta_{disc}$, then $E(u^i) \leq (1 + \gamma)\, \eta_{disc}$.
3. **Efficiency at the stopped iterate.** Assume A1–A5 and $C_2\gamma < 1$. Then $\eta_{disc} \leq C_1(1 + c_b)(E(u^i) + \mathrm{osc}) / (1 - C_2\gamma)$, so at the stopped iterate the bound is two-sided: $E(u^i) \leq \eta_E(u^i) \leq C(1 + c_b)(E(u^i) + \mathrm{osc})$. Here C_1, C_2 and C are independent of ε, σ, h and the iteration index.

Proof.

- **Part 1** is §1 (reliability) applied with the shifted data: the shift is admissible on compatible patches because $\Sigma_a \delta_a \psi_a$ reproduces Res_A on P1 test functions. Part 2 follows from part 1.
- **Part 3, step 1.** Theorem 2 at u^i, with the shifted data in Lemma G, gives an extra term. Write $\|\cdot\|_{*,\omega_a}$ for the local dual norm on the patch ω_a. The extra term is the local dual norm $\|\psi_{def}\|_{*,\omega_a}$ of the shift, so

$$\eta_{disc} \leq C_1(1 + c_b)(E(u^i) + \mathrm{osc}) + C\Big(\Sigma_a \|\psi_{def}\|^2_{*,\omega_a}\Big)^{1/2}.$$

- **Step 2.** By the second inequality of Lemma L, $\Sigma_a \|\psi_{def}\|^2_{*,\omega_a} \leq (d+1)\|\psi_{def}\|^2_*$.
- **Step 3.** $\|\psi_{def}\|_* \leq \eta_{def,A}$, because $\eta_{def,A}$ is a guaranteed upper bound of that dual norm (the construction in `_defect_bound`). Hence the last term is at most $C_2\, \eta_{alg}$.
- **Step 4.** Insert $\eta_{alg} \leq \gamma\, \eta_{disc}$ and absorb. ∎

Hypotheses and sharpness.

- **Part 1 needs nothing.** It holds for every iterate, every solver and every γ. This is the useful part in practice.
- **C_2 is not computable.** "γ small enough" in part 3 is the same unquantified condition as in Ern–Vohralík 2013. The smoke runs suggest $\gamma = 0.1$ is enough.

- **No efficiency of η_alg itself is needed.** Part 3 uses only that η_{def,A} bounds ‖ψ_def‖_* from above. A5, the quasi-optimality of the RT_0 bound, is needed only if one also wants η_alg ≲ ‖ψ_def‖_*, i.e. that η_alg does not stop the iteration later than necessary.
- **Free patches.** On patches with a free (Dirichlet) face, the algebraic residual of a free vertex is absorbed by the patch flux, not by η_alg. This only happens at free vertices next to a Dirichlet face, such as Neumann vertices next to a Dirichlet face. η_disc then contains a small algebraic part, which is harmless for parts 1 and 2.

Literature and novelty.

- **[EV13]** (SISC 35, 2013) is the framework: stopping when η_alg ≤ γ η_disc. There it is built for nonlinear diffusion, with an algebraic flux from ν extra solver steps.
- **Here** the algebraic flux comes from one global RT_0 solve of the spread residual, with no extra iterations. The criterion is stated in the measure E, which is robust in ε.
- **[HPSV21]** (Numer. Math. 147, 2021) proves quasi-optimal cost, but only for strongly monotone Lipschitz operators. The AFC operator is in general not monotone (BJK16 has existence via Brouwer only), so that theory does not apply. Rate-optimal cost for AFC is open.
- **Cheaper algebraic fluxes** exist: the multilevel construction of Papež, Rüde, Vohralík and Wohlmuth (CMAME 2020) is O(N). This would replace the sparse RT_0 solve if that solve becomes the bottleneck in 3D [from memory — verify].
- **AFC-specific stopping.** In my search I found no a posteriori stopping rule for AFC iterations. Ahmed–Jha (arXiv 2602.14504, 2026) relate smoother stabilisations to cheaper iterations; that is based on its abstract, and its full text still has to be checked for a stopping rule.
- **Verdict:** an application of [EV13]. The new parts are the measure E and the iteration-free η_alg.

Numerics (2D smoke runs; BJK, 4 adaptive loops; `stop` = γ):

problem	ε	γ	iterations per loop	E(u^i) at the last loop	η_E/E	η_alg/η_disc at stop
smooth, σ = 0	10^{-2}	— (tol $10^{-8}\sqrt{N}$)	14–26	$1.431\cdot10^{-2}$	3.18–3.46	—
smooth, σ = 0	10^{-2}	0.1	5	$1.392\cdot10^{-2}$	3.29–3.49	0.02–0.04
smooth, σ = 0	10^{-2}	0.01	10	$1.430\cdot10^{-2}$	3.19–3.46	≤ 0.005
smooth, σ = 0	10^{-6}	—	10 000 (not converged)	2.067	3.76–3.99	—
smooth, σ = 0	10^{-6}	0.1	10–15	1.749	3.98–4.53	0.07–0.09
smooth, σ = 0	10^{-6}	0.01	95–1130	2.601	3.81–4.29	≤ 0.009
smooth, σ = 0	10^{-6}	0.001	10 000 (never stops)	2.067	3.76–3.99	0.001–0.004
outflow layers, σ = 0 and 1, BJK and SMUAS	10^{-4}	0.1	5	unchanged	2.8–3.7	≤ 0.03

- **$\varepsilon = 10^{-6}$, $\gamma = 0.1$.** The fixed point for BJK does not converge in 10^4 iterations. The stopped iterate after 10–15 iterations has a smaller E than the 10^4-th iterate, and its bound holds.
- **$\gamma = 10^{-3}$.** η_alg stalls at 0.1–0.4 % of η_disc, so the rule never fires. This is a property of the fixed point, not of the estimator.
- **Effectivities** move by at most 15 %.
- **Tests:** `tests/test_cdr_sv_estimator.py::test_adaptive_stop`, which checks `sv_algebraic` = η_{def,A} to four digits and the stopping rule.

References added.

- [EV13] A. Ern, M. Vohralík, Adaptive inexact Newton methods with a posteriori stopping criteria for nonlinear diffusion PDEs, SIAM J. Sci. Comput. 35 (2013) A1761–A1791. HAL [abstract]
- [HPSV21] A. Haberl, D. Praetorius, S. Schimanko, M. Vohralík, Numer. Math. 147 (2021) 679–725. TU Wien repository [abstract]

- [PRVW20] J. Papež, U. Rüde, M. Vohralík, B. Wohlmuth, Sharp algebraic and total a posteriori error bounds for h and p finite elements via a multilevel approach, CMAME 371 (2020). [from memory — verify]

11. Addendum: variable diffusion ε = ε(x) (gap 7 extension)

Setting.

- ε ∈ L^∞(Ω) with ε ≥ ε₀ > 0, and |||v|||² = ‖ε^{1/2}∇v‖² + σ‖v‖².
- Π₁ε is the cellwise L² projection of ε onto P1. Its cell mean is ε̄_K.
- ε_lo,K and ε_hi,K are the minimum and maximum of ε on K. In the code (`eps_cell_bounds`), ε is fitted by a P1 function on each cell from its values at the degree-4 points. If the fit is exact, ε_lo,K and ε_hi,K are the extreme vertex values of the fit, which are the true extrema; this covers ε piecewise constant or piecewise P1, also discontinuous across cells. On those cells no other samples are mixed in, so a vertex value taken from the other side of a jump cannot make the bound pessimistic. Otherwise they are sampled extrema (cell mean, vertices, degree-4 and degree-6 points), and the estimator reports the number of such cells in `info`.

The construction. The fluxes use D = Π₁ε I + A, a P1 tensor per cell. RTN₁ contains P1 vector fields, so the patch problems stay exact. The remainder is the oscillation η_ε,K = ‖ε^{-1/2}(ε − Π₁ε)∇u_h‖_K.

Lemma V (what P1 tests see). For P1 test functions v, ((ε − Π₁ε)∇u_h, ∇v)_K = (∇u_h·∇v)|_K ∫_K (ε − Π₁ε) = 0, since ∇u_h and ∇v are constant on K and Π₁ε has the cell mean of ε. Three consequences:

- The compatibility Res_A(ψ_a) = 0 at the discrete solution survives with Π₁ε in D. So the defect shift δ_a stays purely algebraic, and `sv_algebraic` (Proposition 5) needs no change.
- For any v ∈ H¹_D, |Res(v) − Res_{Π₁ε}(v)| ≤ Σ_K η_ε,K |||v|||_K. Reliability therefore holds with η_ε,K added cellwise.
- For smooth ε, η_ε,K ≤ C h_K² |ε|_{W^{2,∞}(K)} ε_lo,K^{-1/2} ‖∇u_h‖_K. This is one order above the error, a true oscillation term. A cell-mean ε̄_K would only give O(h)·‖∇ln ε‖_∞ · |||u_h|||_K, which is of the error's own order.

Theorem 6 (variable ε).

1. **Reliability.** Theorem 1 and Corollary 3 hold verbatim, with three changes:
   - η_ε,K is added to the A- and direct-version indicators;
   - in every Poincaré, trace and Friedrichs constant (m_K, c_F, the cut-off, the defect solve), ε is replaced by ε_lo,K;
   - ‖ε^{-1/2}τ‖_K is evaluated with ε pointwise. Nothing beyond ε ∈ L^∞ and ε ≥ ε₀ > 0 is assumed for the bound itself. The computed bound is guaranteed when ε_lo,K and ε_hi,K are the true cell extrema: in the code, when ε is piecewise P1 on the mesh. For other ε they are sampled; the bound then holds up to that sampling, and `info` says so.
2. **Local efficiency.** Theorem 2 holds with constants that also depend on the patch contrast κ_ε(ω̃_K) = max_{ω̃_K} ε / min_{ω̃_K} ε, plus the oscillation η_ε. For ε piecewise constant on the mesh, the within-cell ratio ε_hi,K/ε_lo,K is 1, and only the contrast between cells of a patch remains.

Proof.

- **Part 1.** Lemma V; each Cauchy–Schwarz step weighted by ε pointwise; the inverse inequalities with ε_lo,K, which only make the constants larger.
- **Part 2.** In Lemmas A and G, the patch weight ε_a, the cell weights ε_K and the norms of the competitor are compared using max/min of ε on ω̃_K. In the Verfürth bubble estimates (Lemma D), ε appears only through the local energy norms, at the price of the same ratio. ∎

Minimality and sharpness.

- **Reliability is minimal:** ε ∈ L^∞ with ε ≥ ε₀ is what the norm needs anyway.
- **Efficiency is not robust in jumps as proved.** Removing the κ_ε dependence needs quasi-monotonicity of ε, as in Bernardi–Verfürth 2000 and Petzoldt 2002 for residual estimators and in the Cai et al. equilibrated estimators. It also needs ε-harmonic patch weights in the minimisation.

- **Status in the literature.** I found no conforming equilibrated estimator proved jump-robust without quasi-monotonicity; the Cai–Cai–Zhang conforming results assume it. That is an abstract-level check.
- **Numerics: jumps.** With an aligned jump between two subdomains (a quasi-monotone case), effectivities do not depend on the contrast up to 10^4. Checkerboard (non-quasi-monotone) data are not tested.
- **Numerics: within-cell variation.** ε_lo,K costs a factor (ε_hi,K/ε_lo,K)^{1/2} in the constants. For smooth ε this tends to 1 under refinement.

Literature and novelty.

- **Variable reaction.** Ainsworth–Vejchodský treat non-constant reaction coefficients; robustness in contrast there is in the reaction, not the diffusion [AV14, from memory — verify].
- **Variable diffusion.** ESV10 allow inhomogeneous diffusion in DG, with constants depending on the local contrast.
- **Smoothed fluxes.** Using the L^2 projection $\Pi_1\varepsilon$ in the equilibrated flux, so that the remainder is $O(h^2)$ while the compatibility at P1 tests is kept, is the natural choice. I did not find it stated for algebraic stabilisations.
- **Verdict:** an adaptation, not a new result.

Numerics (2D smoke runs, BJK, uniform meshes n = 8, 16; c = 0 and 1). The table compares the SV estimator with the gap 2 bound `robust` (columns marked "gap 2"); each cell gives the range over the runs.

ε(x)	ε	contrast	η_SV/|||e|||	κ SV/|||e|||_⊕	SV_E/E	gap 2: η_E/E	η_ε/η
smooth, ε·k^{x₁}	10^{-2}	1	1.7–4.6	3.5–4.1	3.0–3.1	3.1–3.9	0
smooth	10^{-2}	10^2	1.1–1.2	3.2–3.4	3.2–3.3	7.1–7.2	0.020–0.036
smooth	10^{-2}	10^4	1.2–1.4	3.5–4.3	3.5–4.3	10.9–12.8	0.071–0.117
smooth	10^{-6}	10^2–10^4	—	4.2–5.5	3.8–4.5	3.8–6.1	≤ 0.006
jump at $x_1 = ½$	10^{-2}	10^2, 10^4	4.0–8.1	3.8–4.5	2.8–3.4	2.9–4.0	0
jump	10^{-6}	10^2, 10^4	—	3.4–5.1	3.4–5.0	3.5–5.1	0
3D smooth (4913 vertices)	10^{-2}	10^2	1.1	3.1	3.0–3.1	6.9	—
3D jump (4913 vertices)	10^{-2}	10^2	3.6–4.1	4.2–4.3	3.2	3.3–4.0	0

- **Guarantee.** All bounds held in every run, in 2D and 3D, and in SV-marked adaptive loops with adaptive stopping (contrast 10^3).
- **Rate of η_ε.** η_ε/η halves from n = 8 to n = 16: it is $O(h^2)$, against an error of $O(h)$.
- **The P1 projection matters.** With the cell mean ε̄_K in the flux (the first version tried), smooth contrast 10^4 gave κ SV/|||e|||_⊕ = 8.3–8.9 and η_ε/η ≈ 0.4.
- **The ABR17-based bounds** (ESV, ABR17, gap 2 `robust`) use the cruder cell-mean version. Their E-effectivity grows to 11–13 at contrast 10^4; the SV estimator is the one to use for variable ε.
- **Tests:** `tests/test_cdr_sv_estimator.py::test_variable_eps` and `::test_eps_cell_bounds`. The problem is `get_problem("variable_eps", kind="smooth" | "jump", contrast=…)`.

References added.

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